

EvidenceTree: Grounded Process Reward Models for Monte Carlo Tree Search over Retrieval Actions in Multimodal Retrieval-Augmented Generation

Anonymous submission

Abstract

Multimodal retrieval-augmented generation (RAG) systems make a sequence of retrieval decisions — what to search, with which modality, and when to stop — yet these decisions are largely unsupervised: outcome-only rewards are sparse and misassign credit, rule-based rewards are noisy, and existing process reward models (PRMs) score *reasoning steps under a fixed context* rather than *retrieval actions that modify the context*. We present EvidenceTree, a multimodal RAG system that performs Monte Carlo Tree Search (MCTS) directly over a typed retrieval action space {TEXT_SEARCH, IMAGE_SEARCH, ANSWER}, guided by a grounded, action-typed PRM trained specifically to evaluate state-modifying retrieval actions. The PRM is supervised with a dual-source step reward that fuses (i) a *local grounding* signal — the relevance of each retrieved result to the question, measured in a single unified CLIP embedding space so that text- and image-side evidence are comparable — and (ii) a *tree-level outcome credit* — the Monte Carlo success rate of all rollouts passing through the node — with an additional *answer-support gate* that prevents parametric lucky guesses from receiving perfect labels. The PRM is trained *generatively* (rationale + score) for dense supervision and queried *score-only* at inference for speed. To train it we construct ETBench-Open, a step-level annotated dataset of retrieval trajectories built entirely on benchmark-provided corpora (InfoSeek, ScienceQA, GQA) with a five-rule rationale quality-control protocol that provably excludes ground-truth leakage. We report the experimental protocol, baselines, and ablation design on InfoSeek, ScienceQA, and GQA; our own results are withheld pending training completion.

1 Introduction

Knowledge-intensive multimodal questions — “When did *this* building officially open?”, asked of a photograph — cannot be answered from model parameters alone: they require retrieving external evidence, often across modalities [8, 16]. Retrieval-augmented generation (RAG) has therefore become the dominant paradigm for knowledge-based visual question

answering (KB-VQA) [7, 23, 39]. However, the *retrieval decisions* of current systems are essentially unsupervised. Fixed retrieve-then-generate pipelines apply the same recipe to every question [11]; reinforcement-learned search agents receive only outcome-level rewards, which are sparse, misassign credit across steps, and are known to induce process hallucination [34, 37]; and imitation-learned decision agents [10] clone corrected search paths without ever learning to *evaluate* a retrieval action — so a single bad retrieval, once taken, silently corrupts everything downstream. Indeed, Chen et al. [10] report that when retrieval goes wrong, 88.6% of their agent’s answers are wrong as well: linear agents cannot recover from a bad search.

Process reward models (PRMs) offer the missing supervision signal: they score intermediate steps rather than final outcomes [22, 33]. But existing PRMs — including multimodal ones [35] and grounded ones [44] — evaluate *reasoning steps under a fixed state*: the image and context are given once and never change. This assumption fails structurally in RAG, where every retrieval action *modifies* the state by appending evidence, where a step’s quality depends on what the retrieval actually returned, and where the system must plan across modalities (search with text, or search with the image?). No existing work simultaneously (A) evaluates the local quality of retrieval actions with grounded signals, (B) supports backtracking when retrieval fails, and (C) plans across retrieval modalities.

EvidenceTree fills this gap with three tightly coupled contributions:

- **An action-typed, grounded PRM for retrieval actions (ET-PRM).** The PRM scores candidate retrieval actions conditioned on the question, the image, and the accumulated evidence bundle. Its training labels fuse a graded *local grounding* signal — computed in one unified CLIP space so text- and image-side results are on a comparable scale — with a *tree-level outcome credit* that estimates each node’s true success potential by Monte Carlo, and an *answer-support gate* that blocks “correct but evidence-free” answers from receiving perfect labels. The PRM is trained *generatively* (rationale + score) for dense supervision and anti-reward-hacking, and queried *score-only* at inference.

- **Retrieval-action MCTS.** We build the search tree directly over retrieval actions: nodes are partial contexts $(q, I, \mathcal{E})$, edges are typed retrieval actions, and selection follows standard UCB1 with the PRM’s value as the sole quality signal. Because the tree branches over *actions*, the system natively supports backtracking out of failed retrieval branches — the failure mode that linear agents cannot escape — and cross-modal planning arises from the PRM’s learned preference for image-based entity disambiguation followed by text-based attribute lookup.
- **ETBench-Open.** A step-level annotated dataset of multimodal retrieval trajectories (target 50–80K trajectories, ~ 300 K step labels) constructed entirely on benchmark-provided corpora — no live web APIs, hence fully reproducible — with a five-rule rationale quality-control protocol that excludes ground-truth leakage by construction. To our knowledge no existing dataset provides step-level supervision for *multimodal retrieval actions*: RAG-ProGuide [43] is text-only, and DeepMMSearchVQA [28] provides trajectories without step-level labels.

We position EvidenceTree against the closest prior systems in Section 2, detail the method in Section 3, describe the dataset construction protocol in Section 4, and lay out the complete experimental design — benchmarks, baselines with their published numbers, metrics, and ablations — in Section 5. Our own numbers are deliberately withheld (marked “–”) pending completion of PRM training; every baseline number in our tables is quoted from the original publication cited in the caption.

2 Related Work

Tree search for multimodal RAG. RCTS [40] performs MCTS over *sets of retrieved examples* to re-rank in-context demonstrations: the search happens after retrieval, over a flat list of already-retrieved pairs, with rule-based answer-matching rewards. AR-MCTS [13] searches in *reasoning space*, using retrieval only to diversify expansion, and trains a PRM over reasoning steps. EvidenceTree differs from both in *where the tree lives*: our nodes are partial retrieval contexts and our edges are retrieval actions, so the search governs the retrieval process itself rather than its aftermath. In text-only RAG, ReasonRAG [43] builds process-supervised preference data via MCTS over an agentic action space, and TreePS-RAG [42] estimates step utility from descendant outcomes; neither is multimodal, neither trains a grounded PRM, and neither searches at inference time with a learned action-value model.

Process reward models. PRMs originated in mathematical reasoning [22, 33]. VisualPRM [35] extends process supervision to multimodal reasoning but scores steps under a *fixed* image context. GroundedPRM [44] verifies each reasoning step with external tools and fuses tool-based local

signals with MCTS-derived outcome signals into rationale-enhanced generative supervision — for mathematical reasoning. We adopt its dual-source philosophy but transplant it to a setting it explicitly does not cover: *state-modifying* retrieval actions, where the object being verified is what an action *returned* rather than a self-contained reasoning statement, where grounding must span modalities in a single comparable space, and where credit must be assigned over a tree of alternative retrievals. Step-level rewards for *search* agents have recently appeared in text-only settings [2, 20, 38]; generative PRMs for web agents [3] score browser actions. None combine multimodal grounding, tree-level credit, and a learned generative PRM for retrieval actions.

Agentic multimodal search. MMSearch-R1 [37] trains an LMM with outcome-only GRPO to invoke image and text search on demand; DeepMMSearch-R1 [28] extends this with query-image cropping and a dedicated VQA dataset; VRAG-RL [34] adds visual perception actions. Closest to our setting, Learning-to-Search (DBAgent) [10] reformulates KB-VQA as a four-action decision process (ANSWER/IMAGE_SEARCH/TEXT_SEARCH/CAPTION) over a curated Wikipedia corpus and reaches state-of-the-art InfoSeek accuracy via supervised fine-tuning on rule-constructed trajectories. All of these are *linear* policies learned by imitation or outcome-only RL: they neither estimate step values nor backtrack. EvidenceTree is complementary — it learns to *evaluate* retrieval actions and to *plan* over them, keeping the policy training-free; DBAgent’s own analysis (29.6% of its samples have correct retrieval but wrong answers, and 88.6% of wrong retrievals end in wrong answers) is direct evidence for the value of step-level evaluation with backtracking.

Step-level datasets. VisualPRM400K [35] contains almost exclusively mathematical reasoning steps; RAG-ProGuide [43] provides 13K process preference pairs for text-only RAG; DeepMMSearchVQA [28] provides multimodal search trajectories without step labels. ETBench-Open is, to our knowledge, the first step-level annotated dataset of *multimodal retrieval actions*, roughly an order of magnitude larger than RAG-ProGuide at target scale.

3 Method

3.1 Problem Formulation

Given a question q and an optional image I , the system interacts with a fixed, benchmark-provided corpus $\mathcal{C}$ over multiple steps. The state after t steps is the partial retrieval context

$$s_t \triangleq (q, I, \mathcal{E}_t), \quad \mathcal{E}_t = \{e_1, \dots, e_{m_t}\}, \quad (1)$$

where $\mathcal{E}_t$ is the *evidence bundle* accumulated so far. At each step the system selects an action $a_t \in \mathcal{A}$; retrieval

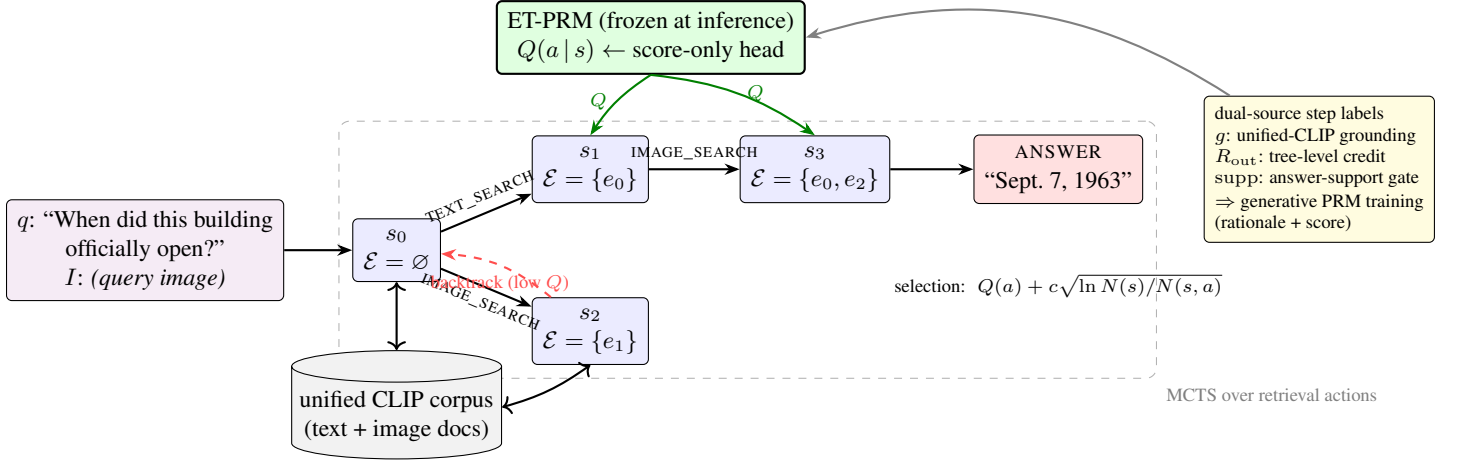


Figure 1: EvidenceTree overview. The search tree is built over *typed retrieval actions*; each edge executes a retrieval against the benchmark-provided corpus (indexed in one unified CLIP space) and appends the results to the evidence bundle $\mathcal{E}$. A grounded, action-typed PRM provides the exploitation term Q of UCB1 selection; low- Q branches are abandoned (backtracking). Training labels (right) fuse local grounding, tree-level Monte Carlo outcome credit, and an answer-support gate, and the PRM is trained generatively on ETBench-Open.

actions modify the state by appending their results, $s_{t+1} = (q, I, \mathcal{E}_t \cup \text{obs}(a_t))$, and the terminal action emits an answer. A complete trajectory is $\tau = (s_0, a_0, s_1, \dots, a_T)$ with $a_T = \text{ANSWER}(y)$. The goal is to choose the sequence of retrieval actions that maximizes the probability that y matches the gold answer y^* , under a fixed rollout budget. Unlike reasoning-step search [13, 35], every non-terminal action here is *state-modifying* and *irreversible within its branch*: evidence, once retrieved, conditions all subsequent decisions on that path. Tree search provides the escape hatch — a branch poisoned by a bad retrieval can be abandoned rather than conditioned on.

3.2 Retrieval Action Space

We type actions by the *modality of the retrieval query*, not of the result:

$$\mathcal{A} = \{ \text{TEXT_SEARCH}(x), \text{IMAGE_SEARCH}(I), \text{ANSWER}(y) \}. \quad (2)$$

Both search actions query the *same* unified index: every corpus document — text chunk or image — is embedded with one CLIP model [30], so a text query or an image query retrieves whichever side of the corpus is most similar. This yields the key multimodal capability: for entity questions where the entity is not named in q (“this bird”, “this building”), IMAGE_SEARCH matches the query image against the corpus to *disambiguate the entity*, after which TEXT_SEARCH retrieves the asked attribute using the evidence-established entity name. We deliberately do not type actions by result modality (a 2×2 split dilutes the visit counts of each branch under small rollout budgets, and the retriever already decides result modality by similarity), do not include compound parallel-search actions (tree expansion already provides multi-candidate exploration, and multi-source fusion is

expressible as a serial trajectory), and do not include state-invariant actions such as captioning (an edge that does not change the observable state would receive no grounding signal and dilute tree-level credit; captioning is folded into the policy prompt instead). Appendix A details these choices.

3.3 Dual-Source Step Reward

Each step of a trajectory receives a graded label in $[0, 1]$ fusing two complementary signals.

Local grounding. Grounding measures *how relevant the results retrieved by this step are to the question* — deliberately scoring the result side rather than the query side, since the query image of IMAGE_SEARCH is fixed by the state and carries no signal about whether the step helped. All results are scored in one CLIP space against the same anchor:

$$g(a_t, s_t) = \max_{r \in \text{obs}(a_t)} \phi_{m(r)} \left(\cos(E_T(q), E_{m(r)}(r)) \right), \quad (3)$$

where E_T and E_I are the CLIP text and image towers, $m(r)$ is the modality of result r , and ϕ_m is a per-modality-pair linear calibration band mapping the raw cosine onto a common $[0, 1]$ relevance scale (same-modality cosines run systematically higher than cross-modal ones). Empty results give $g = 0$. Using a *single* model for both sides is essential: two different encoders would put text- and image-side evidence on different cosine scales, and the PRM would mistake the scale gap for a quality gap, systematically biasing action selection. All grounding scores are graded, not binary — retrieval quality in open corpora is not a 0/1 property (cf. the binary verification of Zhang et al. 44).

Tree-level outcome credit. Outcome credit estimates each node’s true success potential. For a node reached by the action prefix $\pi_t = (a_0, \dots, a_t)$ within one question’s search tree, let $\mathcal{T}(\pi_t)$ denote all complete rollouts passing through it. Then

$$R_{\text{out}}(s_t, a_t) = \frac{1}{|\mathcal{T}(\pi_t)|} \sum_{\tau \in \mathcal{T}(\pi_t)} \mathbb{1}[y(\tau) \approx y^*], \quad (4)$$

a Monte Carlo estimate of the success rate *from this node onward*. This is crucially different from uniformly smearing a trajectory’s final reward over its steps — uniform credit rewards the bad steps of lucky trajectories and punishes the good steps of unlucky ones, whereas Eq. (4) lets two children of the same node receive sharply different labels (cf. shortest-path estimation in Zhang et al. 43; descendant-outcome utilities in Zhang et al. 42). Answer matching $\mathbb{1}[\cdot \approx y^*]$ follows each benchmark’s official protocol (relaxed numeric-range matching for InfoSeek VALUE questions; Appendix C).

Fusion and the answer-support gate. Non-terminal steps fuse the two sources linearly:

$$R(s_t, a_t) = \alpha g(a_t, s_t) + (1 - \alpha) R_{\text{out}}(s_t, a_t), \quad (5)$$

with $\alpha = 0.5$ by default (sensitivity in Section 5.3). Terminal ANSWER steps have no retrieval result to ground, and labeling them by outcome alone creates a subtle poison: a *parametric lucky guess* — an answer that happens to be correct although nothing in the evidence supports it — would receive a perfect label, teaching the PRM that evidence-free answering is optimal. We therefore gate the outcome by an *answer-support* verdict $\text{supp}(y, \mathcal{E}_T) \in [0, 1]$, judged by an LLM over the accumulated evidence (four-way rubric: SUPPORTED/PARTIAL/UNSUPPORTED $\mapsto 1/0.5/0$, plus NOT_REQUIRED $\mapsto$ no gating for questions answerable from the question and image alone by perception or logic — entailment-gating those would systematically punish correct reasoning answers; Appendix B):

$$R_{\text{ans}}(s_T) = R_{\text{out}}(s_T, a_T) \cdot (\gamma + (1 - \gamma) \text{supp}(y, \mathcal{E}_T)), \quad (6)$$

with floor $\gamma = 0.3$. The multiplicative form keeps wrong answers at zero regardless of spurious support; the floor preserves the ordering *wrong* < *correct-but-unsupported* ($\approx \gamma$) < *correct-and-supported* (≈ 1) and buffers verifier noise. Correct-but-unsupported answers are additionally flagged — not discarded — as ready-made rejected sides for preference training (Section 3.4).

3.4 ET-PRM: Architecture and Training

Architecture. ET-PRM is initialized from VisualPRM-8B [35] (itself built on InternVL2.5-8B [9]), inheriting multimodal inductive biases, with an action-conditional value readout. The input serializes the state and the candidate action: question and image, the evidence bundle, and the candidate action’s type and parameters; the output is a rationale followed by a score token.

Generative training, score-only inference. Following the generative reward modeling paradigm [1, 41], ET-PRM is trained to emit a natural-language rationale *and* a score. This is a deliberate choice with three payoffs at our data scale: each sample supervises dozens of tokens rather than one scalar (sample efficiency); the model must articulate *why* an action deserves its score, exposing spurious shortcuts (anti-reward-hacking); and failed training runs can be diagnosed by reading rationales. At inference the rationale is skipped entirely — only the score-token logit is read — so MCTS pays no generation latency.

Three training stages. (1) *SFT*: next-token prediction on ETBench-Open (state, action, rationale, score). (2) *Action-typed preference optimization*: DPO [31] on same-state contrastive pairs mined from the search trees — (successful sibling action, failed action), (TEXT_SEARCH, IMAGE_SEARCH) under the same state, and (supported answer, unsupported-correct answer), the latter using the flagged samples of Eq. (6). Pair construction exploits the same node keys as Eq. (4), so the contrast dimension is “same state, different action” — matching the decision the searcher actually faces — rather than the step-position contrasts used for reasoning PRMs. (3) *Calibration*: temperature/Platt scaling of the score logit on held-out data, so that Q values entering UCB1 have probabilistic meaning.

3.5 PRM-Guided MCTS over Retrieval Actions

Search follows the four canonical MCTS phases with the PRM as the value oracle. *Selection* descends by UCB1 [4, 19]:

$$a^\dagger = \arg \max_a Q(a | s) + c \sqrt{\frac{\ln N(s)}{N(s, a)}}, \quad (7)$$

where Q is the running mean of backed-up rewards seeded by the PRM prior for unvisited edges, and $c = 1.0$ by default. We use *no* additional novelty or modality-coverage bonus: the PRM’s Q is the sole quality signal, and hard-coded exploration preferences would both dilute it and add tuning burden (Appendix A). *Expansion* asks a training-free policy VLM to propose candidate actions (with anti-fabrication and retrieval-chain constraints in the prompt; Appendix E), executes them against the local index, and keeps the top- k by PRM prior. *Simulation* rolls out greedily to a forced answer at depth d_{max} . *Backup* propagates the trajectory reward to the root. Search stops early when the best root path exceeds a quality threshold, or when the rollout budget P is exhausted (defaults: $P=10$, $d_{\text{max}}=3$, $k=3$). Algorithm 1 summarizes the loop.

Algorithm 1 EvidenceTree inference

Require: question q , image I , corpus index, budget P , depth $d_{\max}$

- 1: $\text{root} \leftarrow (q, I, \emptyset)$
- 2: **for** $t = 1 \dots P$ **do**
- 3: $s \leftarrow \text{SELECT}(\text{root})$ by Eq. (7)
- 4: $\mathcal{A}_c \leftarrow$ policy proposals at s ; execute retrievals
- 5: keep top- k children of s by PRM prior $Q(a \mid s)$
- 6: $\tau \leftarrow \text{SIMULATE}$ greedily to ANSWER or $d_{\max}$
- 7: $\text{BACKUP}(\tau, R_{\text{PRM}}(\tau))$
- 8: **if** $\max_{\tau} R_{\text{PRM}}(\tau) > \tau_{\text{stop}}$ **then break**
- 9: **end if**
- 10: **end for**
- 11: **return** answer of the highest-value root path

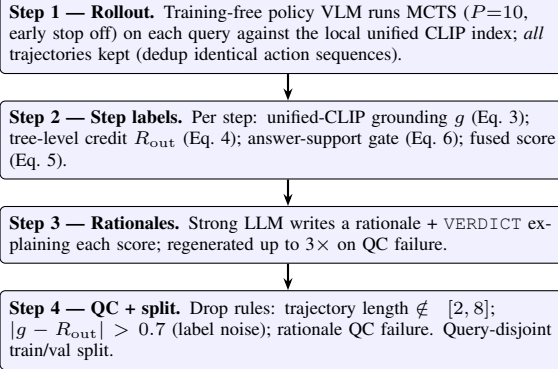


Figure 2: ETBench-Open construction pipeline. Retrieval is fully offline (local indices over benchmark-provided corpora); the only external API is the one-time rationale/support judging, whose outputs ship with the dataset — the construction is reproducible end to end.

4 The ETBench-Open Dataset

ETBench-Open is built by the four-step pipeline of Figure 2, entirely on corpora that ship with the benchmarks: a Wikipedia entity corpus for InfoSeek (chunked full pages of the 1,794 validation entities), the lecture/context corpus for ScienceQA, and scene-graph-derived textual documents paired with images for GQA. This design has three consequences we consider essential for a paper about *decision quality*: reproducibility (no live-web drift), causal cleanliness (performance differences cannot be explained by “the web changed”), and auditability (every evidence item in every label is a corpus document with a stable identifier).

Rationale quality control. Generative PRM training is only as good as its rationales. Each rationale must (a) cite at least one evidence identifier visible in the sample, (b) name the action type, (c) be 30–150 tokens, (d) contain *no ground-truth leakage* — no evaluation phrasing, and no gold-answer string that is absent from the sample’s visible evidence, question, and action input (quoting the answer *from*

cited evidence is legitimate; knowing it without evidence is leakage — rationales must argue *ex-ante*, as at inference the PRM has no oracle), and (e) end with a VERDICT tag that must not contradict the numeric score (preventing say-one-thing-score-another training pairs). Rules (d)/(e) adapt the trajectory-realism constraints of Chen et al. [10] from policy imitation to reward-model supervision.

Scale and status. Target scale is 50–80K trajectories (~300K step labels) across the three benchmarks. Construction statistics of a completed 100-query-per-benchmark pilot — including the step-label and support-verdict distributions that motivated the support gate — are reported in Appendix D. Full-scale construction and PRM training are in progress; accordingly, all EvidenceTree result cells in Section 5 are marked “—”.

5 Experimental Design

5.1 Setup

Benchmarks and metrics. We evaluate on three benchmarks with complementary demands. **InfoSeek** [8]: fine-grained entity knowledge VQA over a Wikipedia corpus; validation split, official relaxed accuracy (string questions: relaxed exact match over the gold set; numeric questions: any predicted number inside the gold range), reported on Unseen-Question / Unseen-Entity / All. **ScienceQA** [27]: multiple-choice science questions; test split accuracy. **GQA** [17]: compositional VQA over real images; test-dev balanced accuracy. ScienceQA and GQA are not knowledge-seeking in the InfoSeek sense; we include them to probe two specific behaviors — whether the PRM learns *when not to retrieve* (ScienceQA reasoning questions) and whether image-query retrieval provides clean perception evidence (GQA scene-graph corpus) — and we analyze them separately from the open-domain headline results.

Implementation. Policy: Qwen2.5-VL-7B-Instruct [6], training-free (prompt in Appendix E). Retrieval: BM25 [32] and dense CLIP indices (FAISS [18]) built offline per benchmark corpus; text top-3, image top-1 following Chen et al. [10]. PRM: VisualPRM-8B fine-tuned with LoRA [15] through the three stages of Section 3.4. Search: $P=10$, $d_{\max}=3$, $k=3$, $c=1.0$, $\alpha=0.5$, $\gamma=0.3$. All results (once reported) will be means over 3 seeds.

5.2 Baselines

We compare against four families: (i) zero-shot MLLMs (no retrieval); (ii) fixed-pipeline multimodal RAG; (iii) reflection- or process-enhanced RAG; (iv) agentic / decision-based search including the strongest published InfoSeek system, DBAgent [10]; plus PRM-with-Best-of- N controls (frozen VisualPRM-8B reranking N sampled trajectories) that isolate

Method	Backbone	Un-Q	Un-E	All
<i>Zero-shot MLLMs (no retrieval)</i>				
BLIP-2 [21]	Flan-T5-XXL	12.7	12.3	12.5
InstructBLIP [12]	Flan-T5-XXL	8.9	7.4	8.1
LLaVA-1.5 [25]	Vicuna-7B	9.6	9.4	9.5
GPT-4V [§]	–	15.0	14.3	14.6
Qwen2.5-VL [§] [6]	7B	22.8	24.1	23.7
<i>Retrieval-augmented (fixed pipeline / reflective)</i>				
Wiki-LLaVA [7]	LLaVA-1.5-7B	30.1	27.8	28.9
PreFLMR [23]	RA-VQAv2	–	–	30.7
EchoSight [39]	LLaMA-3-8B	–	–	31.3
RoRA-VLM [29]	Vicuna-7B	25.1	27.3	–
ReflectiVA [11]	LLaMA-3.1-8B	40.4	39.8	40.1
Wiki-PRF [14]	Qwen2.5-VL-7B	43.3	42.7	42.8
<i>Agentic / decision-based search</i>				
Wiki-R1 [36]	Qwen2.5-VL-7B	47.8	–	44.1
DBAgent [10]	Qwen2.5-VL-7B	46.5	51.0	49.9
VisualPRM-8B + BoN [‡]	Qwen2.5-VL-7B	–	–	–
EvidenceTree (ours)	Qwen2.5-VL-7B	–	–	–

Table 1: InfoSeek (validation, official relaxed accuracy). Baseline numbers are quoted from the cited papers (BLIP-2/InstructBLIP from the official evaluation kit of Chen et al. 8; Wiki-LLaVA and LLaVA-1.5 from Caffagni et al. 7, Table 2; EchoSight from Yan and Xie 39; PreFLMR from Lin et al. 23, Table 7, whose authors note a pre-release InfoSeek setup; RoRA-VLM from Qi et al. 29; ReflectiVA from Cocchi et al. 11; Wiki-PRF from Hong et al. 14; Wiki-R1 from Wiki-R1 authors 36, unseen-entity split not reported; DBAgent (SFT on InfoSeek trajectories) from Chen et al. 10, Table 1. §: zero-shot numbers as compiled in Chen et al. 10, Table 1.) . ‡: control we run under the identical corpus protocol. “–”: training in progress.

the contribution of tree search over sample-and-rerank. Baseline numbers in Tables 1–3 are quoted from the cited publications; rows we will run ourselves under our corpus protocol are marked accordingly.

5.3 Ablation Design

Table 4 pre-registers the ablation grid. **A1** tests whether both reward sources are necessary ($\alpha \in \{0, 0.2, 0.5, 0.8, 1\}$) and whether the support gate matters (Eq. 6 on/off). **A2** varies the exploration constant ($c \in \{0, 0.5, 1, 2\}$; $c=0$ is greedy PRM). **A3** accumulates the PRM training stages (SFT / +DPO / +calibration) against the frozen VisualPRM-8B zero-shot scorer. **A4** grows the action space (TEXT_SEARCH-only $\rightarrow$ +IMAGE_SEARCH). **A5** varies the search budget ($d_{\max} \in \{2, 3, 5\}$, $P \in \{4, 10, 20\}$) against Best-of- N at matched PRM-call budgets — the decisive test that *tree* search, not merely PRM reranking, carries the gains. **A6** evaluates robustness under corpus poisoning [26]: 5–20 adversarial image–text pairs injected per query; we measure the answer flip rate with and without PRM-guided backtracking.

Method	Backbone	Acc.
Human [27]	–	88.40
MM-CoT [45]	UnifiedQA-738M	91.68
LLaVA (fine-tuned) [24]	Vicuna-13B	90.92
Zero-shot [†]	Qwen2-VL-7B	80.33
Zero-shot [†]	InternVL2-8B	93.00
Vanilla RAG [†]	Qwen2-VL-7B	86.68
RCTS [40]	Qwen2-VL-7B	91.44
RCTS [40]	InternVL2-8B	94.20
VisualPRM-8B + BoN [‡]	Qwen2.5-VL-7B	–
EvidenceTree (ours)	Qwen2.5-VL-7B	–

Table 2: ScienceQA (test accuracy). Baseline numbers quoted from the cited papers (MM-CoT from Zhang et al. 45; LLaVA from Liu et al. 24, Table 7; †: as measured in the RCTS paper Yang et al. 40 under its evaluation protocol; RCTS from the same table. AR-MCTS Dong et al. 13 does not evaluate on ScienceQA; methods reporting only the IMG subset, e.g. LLaVA-1.5 at 66.8, are omitted for protocol consistency.) .

Method	Backbone	Acc.
BLIP-2* [21]	Vicuna-13B	41.0
InstructBLIP [12]	Vicuna-7B	49.2
Qwen-VL-Chat [5]	Qwen-7B	57.5
Qwen-VL [5]	Qwen-7B	59.3
LLaVA-1.5 [25]	Vicuna-7B	62.0
LLaVA-1.5 [25]	Vicuna-13B	63.3
EvidenceTree (ours)	Qwen2.5-VL-7B	–

Table 3: GQA (test-dev balanced accuracy). Baseline numbers quoted from the cited papers (Qwen-VL rows from Bai et al. 5; LLaVA-1.5 and InstructBLIP from Liu et al. 25 and Dai et al. 12; *: zero-shot number as reported in Liu et al. 25, Table 1. Existing multimodal RAG and tree-search systems do not report GQA; we include it to probe perception-evidence retrieval over a scene-graph corpus, not as an open-domain RAG battlefield.) .

5.4 Analysis Protocol

Beyond headline accuracy we will report: (i) the retrieval-correctness $\times$ answer-correctness contingency table, whose off-diagonal mass motivates step-level evaluation (cf. 29.6%/88.6% in Chen et al. 10); (ii) PRM score–outcome correlation (Spearman/AUC) against the frozen VisualPRM-8B diagnostic baseline; (iii) per-action-type PRM calibration curves; and (iv) corpus-scale robustness (10K $\rightarrow$ 100K documents).

6 Conclusion

EvidenceTree moves grounded process supervision from fixed-state reasoning to state-modifying retrieval actions: a PRM that evaluates what a retrieval *returned* (in one unified multimodal embedding space), credits each node by the Monte Carlo success of the subtree behind it, refuses to re-

Ablation	Variant	InfoSeek	SQA	GQA
A1 reward	local only ($\alpha=1$)	–	–	–
	outcome only ($\alpha=0$)	–	–	–
	no support gate	–	–	–
A2 search	greedy ($c=0$)	–	–	–
	$c \in \{0.5, 2.0\}$	–	–	–
A3 PRM	frozen VisualPRM	–	–	–
	SFT only	–	–	–
	SFT+DPO	–	–	–
A4 actions	text-only	–	–	–
	+ image search	–	–	–
A5 budget	BoN, matched calls	–	–	–
	$P=4/20$	–	–	–

Table 4: Pre-registered ablation grid (accuracy; “–” = training in progress). A6 (poisoning robustness) reports answer flip rate and is tabulated separately.

ward evidence-free lucky answers, and guides a standard UCB1 tree search that can back out of poisoned branches. The accompanying ETBench-Open provides the first step-level supervision for multimodal retrieval actions, with a leakage-proof rationale protocol. The experimental grid is fully specified; results will populate the “–” cells upon training completion.

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A Action Space: Rejected Designs

Why not a 2×2 (query $\times$ result) action split. Under a roll-out budget of $P=10$, four search actions halve per-branch visit counts; result modality is better decided by the retriever’s similarity in the unified index; and grounding already scores by the *actual* modality of each returned result (Eq. 3), so the action type need not carry it.

Why no PARALLEL_SEARCH. Tree expansion already explores multiple candidates (OR-semantics), and AND-semantics (fuse evidence from two sources) is expressible serially — search A , then B ; the bundle accumulates. A compound action would need bespoke grounding (means over sub-actions), conflict with tree-level credit, and dilute the budget.

Why no CAPTION action. Captioning does not modify the observable state: an edge with an unchanged state has no retrieval result to ground (g undefined), creates parent-child duplicates that dilute Eq. (4), and burns budget. The policy VLM sees the image directly, so caption-style verbalization is folded into the proposal prompt (Appendix E); cf. the caption action of Chen et al. [10], which their inference likewise restricts to a pre-answer query-rewriting aid.

Why no modality-coverage bonus in UCB1. With three actions there is little to “cover”; a λ -weighted novelty term adds three hyperparameters and competes with the PRM’s Q for the same decision, diluting the learned signal. Deferred visual operations (CROP/ZOOM) await a needs analysis of fine-grained visual queries.

B Answer-Support Judging

The judge receives the question, the accumulated evidence, and the proposed answer, and returns one of four verdicts. SUPPORTED: the evidence contains or entails the answer (for multiple choice, entailing the *content* of the chosen option suffices; an absence answer such as “no” is supported when same-image evidence describes the scene and lacks the item). PARTIAL: substantive part unconfirmed. UNSUPPORTED: the answer rests on external facts absent from the evidence. NOT_REQUIRED: derivable from the question and visible image alone by perception/logic/commonsense — never applicable to fine-grained entity facts (species data, dates, measurements). The fourth verdict maps to *no gating* (outcome-only), which repairs the systematic penalty that entailment-gating inflicts on reasoning-type multiple-choice answers, while leaving the anti-guess gate fully active on knowledge-seeking questions. Raw verdicts ship with the dataset for auditability. An offline lexical fallback (answer content-word/number coverage in the evidence, with unmatched numbers vetoing) supports API-free smoke runs; unparseable or

unverifiable cases return no verdict and fall back to outcome-only rather than fabricating a value.

C Dataset Schema and Quality Control

Each released sample carries: identifiers (sample/trajectory/query); the state snapshot before the action (actions so far, evidence bundle); the typed action; the observed results; labels g, R_{out} (with the count of trajectories through the node), supp with its raw verdict, the fused score; the rationale, its verdict tag, QC status; and generation provenance (policy model and deployment, retriever, MCTS configuration, seeds, pipeline commit). Hard drop rules (the only ways a sample is excluded): trajectory length outside $[2, 8]$; non-answer steps with $|g - R_{\text{out}}| > 0.7$ (label self-contradiction); rationale QC failure after three regenerations. Failed trajectories are retained — their low-scored steps are the PRM’s negative curriculum — and correct-but-unsupported answers are retained with flags as DPO rejected candidates. Validation enforces score-fusion identities, per-field ranges, the answer $\leftrightarrow$ null-grounding rule, query-disjoint splits, and redistributable image references.

D Construction Pilot Statistics

A 100-query-per-benchmark construction pilot ($P=10$) produced 207 / 170 / 166 deduplicated trajectories and 464 / 366 / 355 step samples on InfoSeek / ScienceQA / GQA respectively, of which 426 / 328 / 341 survived QC (drops dominated by the grounding–outcome gap rule). The pilot surfaced — and motivated the fixes for — three construction-level failure modes: (i) *parametric lucky guesses* on InfoSeek (answers judged correct with zero evidence support), addressed by the support gate of Eq. (6); (ii) *systematic entailment penalties* on ScienceQA reasoning questions (105 of 170 correct answers initially judged unsupported), addressed by the NOT_REQUIRED verdict; and (iii) *action imbalance and trajectory collapse* on InfoSeek (image:text search ratio 166:54 against a text-only corpus; ≈ 2 deduplicated trajectories per query), addressed by retrieval-chain and proposal-diversity constraints in the policy prompt (Appendix E). We report these numbers for transparency: dataset papers rarely publish their construction pathologies, yet they determine label quality.

E Prompts

Policy proposal prompt (excerpt). The policy must not invent entity identities not established by the image or evidence; an entity name *that already appears in the evidence* is explicitly not a guess and should seed TEXT_SEARCH; once the entity is identified but the asked attribute is missing, the next

proposal must pair the evidence-established name with the attribute rather than answering from an entity match alone; candidate queries within one proposal must differ in angle (entity+attribute / bare keywords / rephrasing) and never repeat a query already executed on the path; answering is licensed only when the evidence supports it or the question is answerable from the image and question alone.

Rationale prompt (excerpt). The writer judges the step only from the question, evidence, and action — as if the final outcome were unknown; never mentions gold answers or evaluation language; states plainly when evidence fails to support the step; and ends with VERDICT: good|mixed|poor consistent with the assigned score.

F Notation

Symbol	Meaning
q, I	question; optional query image
$s_t, \mathcal{E}_t$	state; evidence bundle (Eq. 1)
$\mathcal{A}$	typed action space (Eq. 2)
$\tau, y(\tau), y^*$	trajectory; its answer; gold answer
$g(a, s)$	local grounding (Eq. 3)
$R_{\text{out}}(s, a)$	tree-level outcome credit (Eq. 4)
$\text{supp}(y, \mathcal{E})$	answer-support verdict (Eq. 6)
α, γ	fusion weight (0.5); support floor (0.3)
$Q(a s)$	PRM action value (exploitation term)
$N(s), N(s, a)$	node / edge visit counts
c	UCB1 exploration constant (1.0)
P, d_{max}, k	rollouts (10); max depth (3); children (3)
τ_{stop}	early-stop threshold

Table 5: Notation.